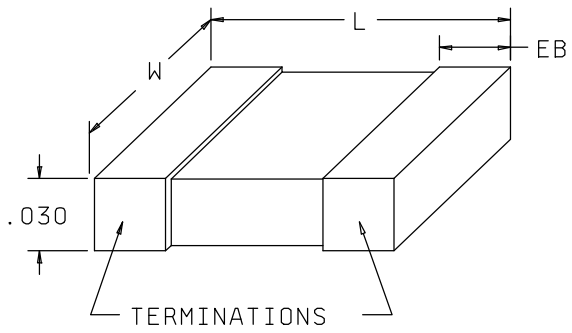
INCH-POUND
MIL-PRF-55681/6A
3 July 1997
SUPERSEDING
MIL-C-55681/6
30 July 1981

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, MULTIPLE LAYER, FIXED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, STYLES CDR26, CDR27, CDR28, CDR29, AND CDR30

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55681.



Inches	mm
.010	0.25
.030	0.76
.040	1.02

Dimensions				
Style	L	W	EB	
			min	max
CDR26	.150 (3.81) ±.015 (0.38)	.150 (3.81) ±.015 (0.38)	.010	.030
CDR27	.190 (4.83) ±. .019 (0.48)	.190 (4.83) ±. .019 (0.48)	.010	.030
CDR28	.330 (8.38) ±. .033 (0.84)	.330 (8.38) ±. .033 (0.84)	.010	.040
CDR29	.400 (10.16) ±. .040 (1.02)	.400 (10.16) ± .040 (1.02)	.010	.040
CDR30	.540 (13.72) ±. .054 (1.37)	.400 (10.16) ± .040 (1.02)	.010	.040

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimensions and tolerances are for bare chip. For solder coated terminations, add .020 (0.51 mm) to the length and .012 (0.30 mm) to the width and thickness.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: F = ±1 percent; G = ±2 percent; J = ±5 percent; K = ±10 percent; M = ±20 percent. See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Product level: Non-established Reliability - C; Established Reliability - M, P, R, and S.

Marking: In accordance with MIL-PRF-55681.

Voltage conditioning: In accordance with MIL-PRF-55681 except that 100 percent of rated voltage shall be applied to the unit at the maximum rated temperature +4°C, -0°C for 100 ±4 hours.

Insulation resistance: In accordance with MIL-PRF-55681 except that for the test conditions, 500 volts shall be applied through a resistor sufficient to limit the charging current to a maximum of 30 mA.

Dielectric withstanding voltage: In accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

a. 150 percent of dc rated voltage shall be applied for 1,000 volts, and 120 percent of dc rated voltage shall be applied for 2,000 volts or higher. It is recommended that the capacitors be immersed in an inert medium to prevent arcing or higher leakage currents.

b. Duration of application of test voltage: 1 second minimum. The test voltage shall be raised from zero to the specified value within 1 minute maximum.

c. Points of application of test voltage: Between the capacitor terminals.

d. Limiting value of surge current: 10 mA maximum.

e. Examination after test: Capacitors shall be examined for evidence of damage and breakdown.

Voltage-temperature limits: In accordance with MIL-PRF-55681 except that for capacitors with the BZ rated temperature and voltage temperature limits, 60 percent of rated voltage shall be applied in steps E through G of the voltage-temperature limit cycle table. Capacitors shall be kept at each temperature until temperature equilibrium is attained.

Life: In accordance with MIL-PRF-55681 except that for the operating conditions, capacitors shall be subjected to 100 percent of rated voltage. The surge current shall not exceed 50 mA. When necessary, a suitable current-limiting resistor shall be inserted into the current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics.

Part or Identifying Number (PIN) 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits
1,000 V dc			
CDR26BP681F - - -	680	F, G, J	BP
CDR26BP821F - - -	820	F, G, J	BP
CDR26BP102F - - -	1,000	F, G, J	BP
CDR26BP122F - - -	1,200	F, G, J	BP
CDR26BP152F - - -	1,500	F, G, J	BP
CDR26BP182F - - -	1,800	F, G, J	BP
CDR26BP222F - - -	2,200	F, G, J	BP
CDR27BP272F - - -	2,700	F, G, J	BP
CDR27BP332F - - -	3,300	F, G, J	BP
CDR27BP392F - - -	3,900	F, G, J	BP
CDR27BP472F - - -	4,700	F, G, J	BP
CDR28BP562F - - -	5,600	F, G, J	BP
CDR28BP682F - - -	6,800	F, G, J	BP
CDR28BP822F - - -	8,200	F, G, J	BP
CDR28BP103F - - -	10,000	F, G, J	BP
CDR28BP123F - - -	12,000	F, G, J	BP
CDR29BP153F - - -	15,000	F, G, J	BP
CDR29BP223F - - -	22,000	F, G, J	BP
CDR30BP273F - - -	27,000	F, G, J	BP
CDR30BP333F - - -	33,000	F, G, J	BP
CDR26BX272F - - -	2,700	K, M	BX
CDR26BX332F - - -	3,300	K, M	BX
CDR26BX392F - - -	3,900	K, M	BX
CDR26BX472F - - -	4,700	K, M	BX
CDR26BX562F - - -	5,600	K, M	BX
CDR26BR682F - - - 2/	6,800	K, M	BR
CDR26BR822F - - - 2/	8,200	K, M	BR
CDR26BZ103F - - - 3/	10,000	K, M	BZ
CDR27BR123F - - - 2/	12,000	K, M	BR
CDR27BR153F - - - 2/	15,000	K, M	BR
CDR27BZ223F - - - 3/	22,000	K, M	BZ
CDR28BX273F - - -	27,000	K, M	BX
CDR28BX333F - - -	33,000	K, M	BX
CDR28BX393F - - -	39,000	K, M	BX
CDR28BR473F - - - 2/	47,000	K, M	BR
CDR29BX563F - - -	56,000	K, M	BX
CDR29BR823F - - - 2/	82,000	K, M	BR
CDR30BX104F - - -	100,000	K, M	BX
CDR30BR124F - - - 2/	120,000	K, M	BR
CDR30BZ154F - - - 3/	150,000	K, M	BZ

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits
2,000 V dc			
CDR26BP331G - - -	330	F, G, J	BP
CDR26BP391G - - -	390	F, G, J	BP
CDR26BP471G - - -	470	F, G, J	BP
CDR26BP561G - - -	560	F, G, J	BP
CDR27BP681G - - -	680	F, G, J	BP
CDR27BP821G - - -	820	F, G, J	BP
CDR27BP102G - - -	1,000	F, G, J	BP
CDR27BP122G - - -	1,200	F, G, J	BP
CDR28BP152G - - -	1,500	F, G, J	BP
CDR28BP182G - - -	1,800	F, G, J	BP
CDR28BP222G - - -	2,200	F, G, J	BP
CDR28BP272G - - -	2,700	F, G, J	BP
CDR29BP332G - - -	3,300	F, G, J	BP
CDR29BP392G - - -	3,900	F, G, J	BP
CDR29BP472G - - -	4,700	F, G, J	BP
CDR29BP562G - - -	5,600	F, G, J	BP
CDR30BP682G - - -	6,800	F, G, J	BP
CDR30BP822G - - -	8,200	F, G, J	BP
CDR26BX102G - - -	1,000	K, M	BX
CDR26BR122G - - - 2/	1,200	K, M	BR
CDR26BR152G - - - 2/	1,500	K, M	BR
CDR26BR182G - - - 2/	1,800	K, M	BR
CDR26BR222G - - - 2/	2,200	K, M	BR
CDR27BR272G - - - 2/	2,700	K, M	BR
CDR27BR332G - - - 2/	3,300	K, M	BR
CDR27BZ392G - - - 3/	3,900	K, M	BZ
CDR27BX472G - - -	4,700	K, M	BX
CDR27BX562G - - -	5,600	K, M	BX
CDR28BX682G - - -	6,800	K, M	BX
CDR28BX822G - - -	8,200	K, M	BX
CDR28BR103G - - - 2/	10,000	K, M	BR
CDR28BR123G - - - 2/	12,000	K, M	BR
CDR29BX153G - - -	15,000	K, M	BX
CDR29BR223G - - - 2/	22,000	K, M	BR
CDR30BR273G - - - 2/	27,000	K, M	BR
CDR30BZ333G - - - 3/	33,000	K, M	BZ

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits
3,000 V dc			
CDR26BP221H - - -	220	F, G, J	BP
CDR26BP271H - - -	270	F, G, J	BP
CDR27BP391H - - -	390	F, G, J	BP
CDR27BP471H - - -	470	F, G, J	BP
CDR27BP561H - - -	560	F, G, J	BP
CDR28BP821H - - -	820	F, G, J	BP
CDR28BP102H - - -	1,000	F, G, J	BP
CDR28BP122H - - -	1,200	F, G, J	BP
CDR29BP152H - - -	1,500	F, G, J	BP
CDR29BP182H - - -	1,800	F, G, J	BP
CDR29BP222H - - -	2,200	F, G, J	BP
CDR30BP272H - - -	2,700	F, G, J	BP
CDR30BP332H - - -	3,300	F, G, J	BP
CDR26BR561H - - - 2/	560	K, M	BR
CDR26BR681H - - - 2/	680	K, M	BR
CDR26BZ821H - - - 3/	820	K, M	BZ
CDR27BZ152H - - - 3/	1,500	K, M	BZ
CDR27BZ182H - - - 3/	1,800	K, M	BZ
CDR28BX272H - - -	2,700	K, M	BX
CDR28BX332H - - -	3,300	K, M	BX
CDR28BX392H - - -	3,900	K, M	BX
CDR29BX472H - - -	4,700	K, M	BX
CDR29BX562H - - -	5,600	K, M	BX
CDR30BX682H - - -	6,800	K, M	BX
CDR30BR103H - - - 2/	10,000	K, M	BR
CDR30BR123H - - - 2/	12,000	K, M	BR

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits
4,000 V dc			
CDR26BP151J - - -	150	F, G, J	BP
CDR27BP271J - - -	270	F, G, J	BP
CDR27BP331J - - -	330	F, G, J	BP
CDR28BP391J - - -	390	F, G, J	BP
CDR28BP471J - - -	470	F, G, J	BP
CDR28BP561J - - -	560	F, G, J	BP
CDR28BP681J - - -	680	F, G, J	BP
CDR29BP821J - - -	820	F, G, J	BP
CDR29BP102J - - -	1,000	F, G, J	BP
CDR29BP122J - - -	1,200	F, G, J	BP
CDR30BP152J - - -	1,500	F, G, J	BP
CDR30BP182J - - -	1,800	F, G, J	BP
CDR30BP222J - - -	2,200	F, G, J	BP
CDR26BX151J - - -	150	K, M	BX
CDR26BX181J - - -	180	K, M	BX
CDR26BX221J - - -	220	K, M	BX
CDR26BX271J - - -	270	K, M	BX
CDR26BX331J - - -	330	K, M	BX
CDR26BX391J - - -	390	K, M	BX
CDR26BX471J - - -	470	K, M	BX
CDR27BX561J - - -	560	K, M	BX
CDR27BX681J - - -	680	K, M	BX
CDR27BX821J - - -	820	K, M	BX
CDR27BX102J - - -	1,000	K, M	BX
CDR27BX122J - - -	1,200	K, M	BX
CDR28BX152J - - -	1,500	K, M	BX
CDR28BX182J - - -	1,800	K, M	BX
CDR28BR222J - - - 2/	2,200	K, M	BR
CDR29BX272J - - -	2,700	K, M	BX
CDR29BR332J - - - 2/	3,300	K, M	BR
CDR29BR392J - - - 2/	3,900	K, M	BR

1/ Complete PIN shall include additional symbols to indicate capacitance tolerance, termination finish, and product level.

2/ With respect to rated temperature and voltage-temperature limits in the original issue of this specification sheet: BR is interchangeable with BX. These parts shall be stocked under the same applicable NSN's.

3/ With respect to rated temperature and voltage-temperature limits in the original issue of this specification sheet: BZ is interchangeable with BX. These parts shall be stocked under the same applicable NSN's.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1923-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.